

ECON 312 — Intermediate Microeconomics

Lecture Notes — Ch. 5: Choice

Fall 2026

We're going to put the budget and utility pieces together now to understand what a consumer *chooses* and how that choice responds to changes in prices and income.

Outline

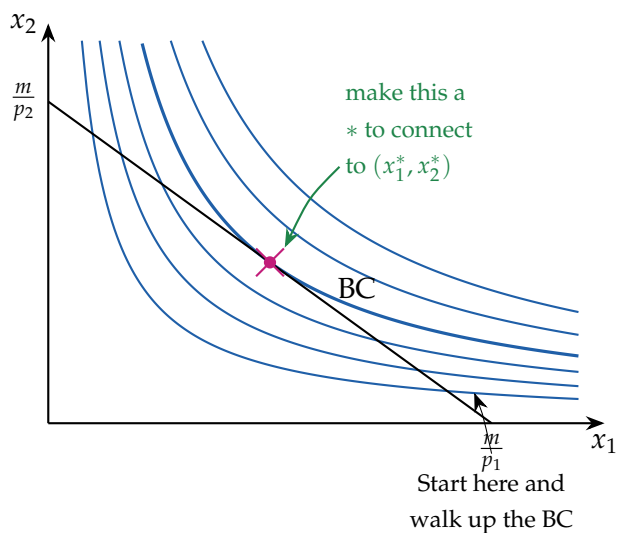
- ① What optimal choices look like.
- ② How to find (using ~~hard~~ fun math) optimal choices.
- ③ Using optimal choices to do a welfare analysis.

Remember

The consumer chooses the *most-preferred* bundle from their consumption set!

① What Optimal Choices Look Like

Let's visualize the problem.



→ Initially, each step brings you to a more-preferred bundle b/c you are going up an indifference curve.

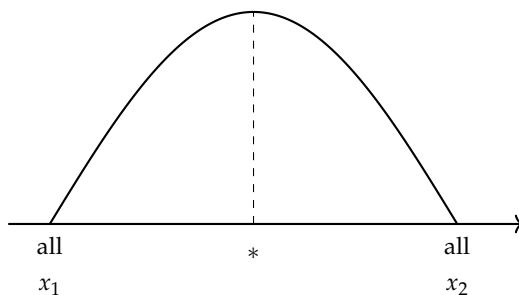
Q Why? B/c the BC crosses an indifference curve.

How do things change when we get to *?

→ We're still crossing indifference curves, but now in the *other* direction!

↪ This means we are getting worse, *not* better.

So utility moving along the budget constraint looks like:



So * is the most-preferred bundle.

◇ It is also where we switched the direction we crossed the indifference curve from.

↪ We didn't actually cross an indifference curve, we just "kissed" it.

(x_1^*, x_2^*) is where the BC is tangent to the indifference curve!

↪ It is where $-\frac{p_1}{p_2} = \text{MRS}$!

What just happened? We went looking for the solution of the consumer optimization problem, and it just so happened that the solution *exactly* lined up with the very object I've been priming you to think about!

But this solution actually makes total sense if you think about it!

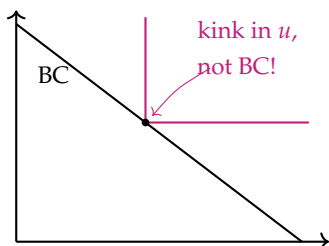
→ The market prices p_1 & p_2 are a "machine" that converts 1 x_1 into $\frac{p_1}{p_2} x_2 \dots$

→ If (x_1^*, x_2^*) is not where the MRS is, then the consumer can easily increase their utility by trading!

Q When does tangency fail?

But is this always true? Could there ever be a time when this did not work? What are the possibilities?

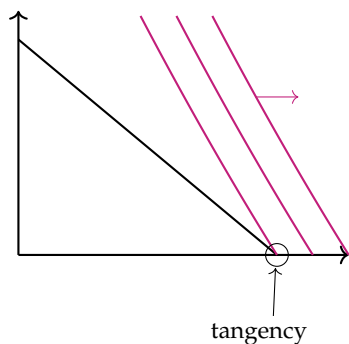
A A technical work-around.



→ Grinchy: "A tangent doesn't "technically" exist!"

+ Yah, but just extend "tangent" in the right way and it works again...

B A corner solution (boundary optimum).

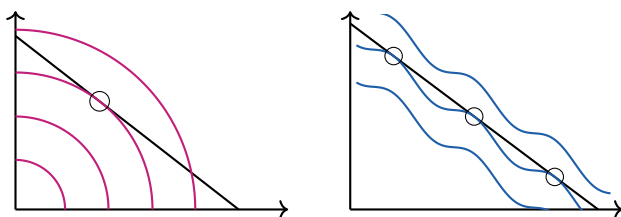


A constraint problem!

+ But if we could extend to allow for negative x_2 , we would get around this.

→ Keep this on the back of our minds...

© Multiple / flipped tangency.



A sign problem.

→ If the tangency is from bending backwards (concavity), we run into the same issue.

★ ⇒ Require *convex* preferences.

The Demanded Bundle

Before diving into finding the most-preferred bundle, let's give it a name: the consumer's demanded bundle. We often want to think of it as a function:

$$x_1(p_1, p_2, m) = \text{the } x_1^* \text{ of the problem w/ } p_1, p_2, \text{ and } m.$$

Likewise, $x_2(p_1, p_2, m)$. These are the demand functions — also called (x_1^*, x_2^*) .

⇒ Our running goal is to study these functions.

Q Keep asking

How does the optimal bundle change with p_1, p_2, m ?

② Actually Computing x^*

Setup: let's have Cobb–Douglas utility

$$u(x_1, x_2) = x_1^c x_2^d \quad \text{w/} \quad p_1 x_1 + p_2 x_2 = m.$$

Change $\leq$ to $=$ b/c will spend all \$\$\$

Problem: find (x_1^*, x_2^*) that maximizes $u(x_1^*, x_2^*)$.

*** We are going to take 3 different angles/approaches ***

Method ①: MRS Condition + Budget Exhaustion

By previous material, we know that

$$\underline{a} \quad \frac{\partial u(x_1, x_2)/\partial x_1}{\partial u(x_1, x_2)/\partial x_2} = \frac{p_1}{p_2}, \quad \& \quad \underline{b} \quad p_1 x_1 + p_2 x_2 = m.$$

$\Rightarrow$ 2 equations, 2 unknowns. Let's do it!

Remember to emphasize

Remember to emphasize the difference b/w $\frac{\partial}{\partial x_1}$ & $\frac{d}{dx_1}$.

The partials:

$$\frac{\partial u(x_1, x_2)}{\partial x_1} = \frac{\partial x_1^c x_2^d}{\partial x_1} = c x_1^{c-1} x_2^d, \quad \frac{\partial x_1^c x_2^d}{\partial x_2} = d x_1^c x_2^{d-1}.$$

Use $\underline{a}$:

$$\frac{c x_1^{c-1} x_2^d}{d x_1^c x_2^{d-1}} = \frac{c}{d} \cdot \frac{x_2}{x_1} = \frac{p_1}{p_2}.$$

Now solve $\underline{b}$ for x_1 :

$$p_1 x_1 + p_2 x_2 = m \Rightarrow x_1 = \frac{m - p_2 x_2}{p_1}.$$

Substitute:

$$\frac{c}{d} \cdot \frac{x_2}{\frac{m - p_2 x_2}{p_1}} = \frac{p_1}{p_2} \Rightarrow p_2 c x_2 = d(m - p_2 x_2) \Rightarrow p_2 c x_2 = d m - d p_2 x_2.$$

$$x_2^* = \frac{d m}{p_2(c + d)} = \frac{d}{c + d} \cdot \frac{m}{p_2} \quad (\text{pretty version})$$

Repeat for x_1 , get

$$x_1^* = \frac{c}{c + d} \cdot \frac{m}{p_1}$$

Method ②: Maximization Problem + Substitution

But what if that seemed... too easy?

$$\max x_1^c x_2^d \quad \text{s.t.} \quad p_1 x_1 + p_2 x_2 = m.$$

$\Rightarrow$ Get rid of the constraint by (again)

$$x_1(x_2) = \frac{m - p_2 x_2}{p_1} \quad (\text{same as before}).$$

$\Rightarrow$ the problem becomes

$$\max \left(\frac{m - p_2 x_2}{p_1} \right)^c x_2^d.$$

Now take the total derivative

Critical! Same as partial b/c only one variable

$$\begin{aligned}\frac{d}{dx_2} u(x_1(x_2), x_2) &= c \left(\frac{m - p_2 x_2}{p_1} \right)^{c-1} \frac{(-p_2)}{p_1} x_2^d + \left(\frac{m - p_2 x_2}{p_1} \right)^c d x_2^{d-1} = 0. \\ \Rightarrow \left(\frac{m - p_2 x_2}{p_1} \right)^c d x_2^{d-1} &= c \left(\frac{m - p_2 x_2}{p_1} \right)^{c-1} \frac{p_2}{p_1} x_2^d.\end{aligned}$$

Cancel terms:

$$d \left(\frac{m - p_2 x_2}{p_1} \right) = c \frac{p_2}{p_1} x_2 \Rightarrow \boxed{x_2^* = \frac{d}{c+d} \cdot \frac{m}{p_2}} \checkmark$$

Follow your nose to get x_1 in the same way.

Aside: why does this work? We did $\frac{d}{dx_2} u(x_1(x_2), x_2) = 0$, which is just (chain rule)

$$\frac{\partial u(x_1(x_2), x_2)}{\partial x_1} \cdot \frac{\partial x_1(x_2)}{\partial x_2} + \frac{\partial u(x_1(x_2), x_2)}{\partial x_2} = 0,$$

$$\text{but } \frac{\partial x_1(x_2)}{\partial x_2} = \frac{\partial}{\partial x_2} \left(\frac{m - p_2 x_2}{p_1} \right) = -\frac{p_2}{p_1}. \Rightarrow \text{w/ algebra...}$$

$$\frac{\partial u(x_1(x_2), x_2) / \partial x_1}{\partial u(x_1(x_2), x_2) / \partial x_2} = \frac{p_1}{p_2} \dots$$

So we just reused / rederived the MRS condition...

Method ③: Solve the Lagrangian

But we have a 3rd method!

Previously, we used substitution to get rid of the constraint. But that may not always be easy. This method handles this well.

1st, write the BC as an “= 0” constraint:

$$\underline{a} \quad p_1 x_1 + p_2 x_2 - m = 0.$$

2nd, write a new optimization problem:

$$\max x_1^c x_2^d - \underset{\text{Lambda, for Lagrange}}{\lambda} (p_1 x_1 + p_2 x_2 - m) \quad (= \mathcal{L}).$$

Why would you do this?

$\Rightarrow$ Think about tangency and $\lambda \rightarrow$ when you have $\frac{\partial}{\partial \lambda} (\mathcal{L}) = 0$, it follows that $p_1 x_1 + p_2 x_2 - m = 0 \rightarrow$ so the BC is satisfied.

Now, you have an unconstrained maximization problem, and you only need to find the solution to the partials:

$$\frac{\partial \mathcal{L}}{\partial x_1} = 0, \quad \frac{\partial \mathcal{L}}{\partial x_2} = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 0.$$

It's like magic, except it works every time. We get

$$\begin{array}{ll} \underline{a} & c x_1^{c-1} x_2^d - \lambda p_1 = 0 \\ \underline{b} & d x_1^c x_2^{d-1} - \lambda p_2 = 0 \\ \underline{c} & p_1 x_1 + p_2 x_2 - m = 0 \end{array} \implies \text{divide } \underline{a} \text{ by } \underline{b}: \frac{c x_2}{d x_1} = \frac{p_1}{p_2}$$

...which is just the MRS condition again!

Lagrange is overkill here, but we will use it numerous times in the future.

③ What Else Can the MRS Help Us Understand?

A: The trade-off that everyone is willing to make!

- In a well-functioning market, all consumers face the same prices.
- ⇒ MRS is the same for every consumer.
- ⇒ New production technologies only help if they exchange goods at a better rate.

B: Welfare implications of various taxes.

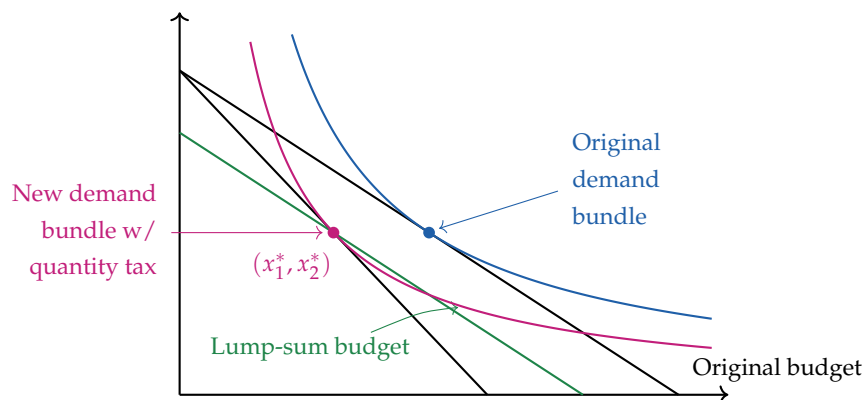
Let's consider two different kinds of taxes and ask which one they would prefer (neither, obviously).

Tax 1 — Quantity tax: raise price of good 1 by t .

Q What is the new problem?

$$\max u(x_1, x_2) \quad \text{s.t.} \quad (p_1 + t)x_1 + p_2 x_2 \leq m.$$

Q Graphically, how does this look?



Q Revenue raised? $x_1^* \cdot t \dots$

What if we just raised this revenue directly by imposing a tax of $x_1^* \cdot t$?

Q What is the new problem?

$$\max u(x_1, x_2) \quad \text{s.t.} \quad p_1 x_1 + p_2 x_2 \leq m - \underset{\text{fixed}}{x_1^*} t.$$

⇒ Has the same slope as before, but shifted in.

⇒ Old point (x_1^*, x_2^*) is still attainable, so we are good there & have an intercept.

Q *So which one does the consumer prefer?*

⇒ **Lump-sum tax.**

Q *Why?*

→ The quantity tax distorts the prices and causes them to substitute away (artificially) from x_1 .